

Essential Summary

A Modified Generalized Harmonic Function Perturbation Method and Its Application in Analyzing Generalized Duffing–Harmonic –Rayleigh–Liénard Oscillator

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1 Background and Motivation

This is the **foundational paper** of the MGHFP method series. Many classical analytical methods (perturbation-based, energy balance-based, homotopy-based) can analyze nonlinear oscillators quantitatively — but only when the restoring force and damping are **simple**. Once the oscillator becomes complicated (e.g., with mixed polynomial and irrational nonlinearities), these methods **cannot execute their procedures symbolically** without first assigning numerical values to all system parameters. This means they can only produce isolated solution snapshots in parameter space, never a continuous, analytical picture of how solutions evolve with changing parameters.

This work introduces the **modified generalized harmonic function perturbation (MGHFP) method** to overcome this limitation. The core innovation is the introduction of the **composite Simpson quadrature formula** into the classical GHFP framework, enabling the entire perturbation procedure to proceed purely symbolically — no parameter pre-assignment required. This is the foundation upon which all subsequent MGHFP applications (IJNLM 2025, Phys. Scr. 2026) are built.

2 System Model

The generalized Duffing–Harmonic–Rayleigh–Liénard (DHRL) oscillator:

$$\ddot{x} + \mu x + \lambda x^3 + \nu \frac{x}{1 - x^2} = \varepsilon(\mu_c + \mu_2 x^2 + \mu_4 x^4 + \mu_6 x^6 + \mu_{22} x \dot{x}^2 + \mu_{24} x^3 \dot{x}^2) \dot{x} \quad (1)$$

where λ , μ , ν are stiffness parameters (Duffing, linear, Harmonic), and the right side contains a **6-term generalized nonlinear damping** combining Rayleigh-type ($\dot{x}^2$ terms) and Liénard-type (x^2, x^4, x^6 terms) damping. This oscillator is deliberately chosen as a maximally challenging test case — its restoring force mixes polynomial and irrational nonlinearities, and its damping spans multiple functional forms.

3 The MGHFP Method — Key Innovation

3.1 Classical GHFP Limitation

The classical GHFP method proceeds via: (1) nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$, (2) Fourier expansion of the solution, (3) perturbation expansion in ε , and (4) evaluation of integrals involving the nonlinear terms to obtain Fourier coefficients. Step (4) is the bottleneck — for complicated oscillators, these integrals **cannot be evaluated analytically**. The classical GHFP therefore

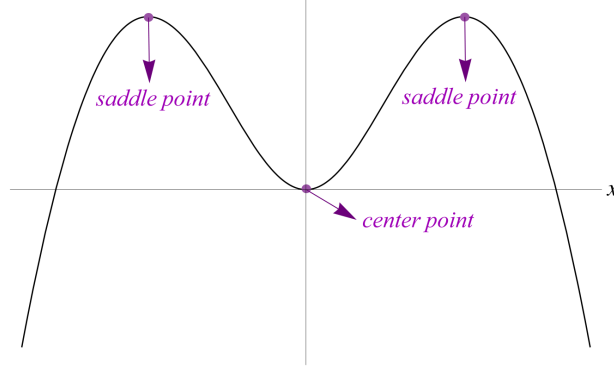


Figure 1: **Figure 6.** Potential portrait of the DHRL oscillator. The rich potential landscape enables homoclinic and heteroclinic bifurcations, making this an ideal testbed for the MGHFP method.

requires all system parameters to be assigned numerical values *before* the integrals can be computed numerically. The result is semi-analytic: valid only for that specific parameter set.

3.2 The MGHFP Fix

The MGHFP method introduces the composite Simpson quadrature formula to **discretize** the integrals in Step (4):

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(x_n) \right] \quad (2)$$

By dividing the integration interval into eight equal subintervals and applying this formula, the Fourier coefficients p_{2i} are expressed as **explicit algebraic functions** of a_0 , b , and the system parameters. The entire perturbation procedure now runs purely symbolically.

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

The analytical relationship between limit cycle amplitude A and control parameter μ_c :

$$\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)} \left(E_2\mu_2 + E_4\mu_4 + E_6\mu_6 + E_{22}\mu_{22} + E_{24}\mu_{24} \right) \quad (3)$$

where E_i are functions of a_0 , b , and the Fourier coefficients p_{2i} computed via composite Simpson integration. Given parameters, the limit cycle amplitude is obtained **analytically**.

4.2 Stability Criterion

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu_c + \sum \mu_i (a_0 \cos^2 \varphi + b)^i + \sum \mu_{ij} (a_0 \cos^2 \varphi + b)^i (\dot{x}_0)^2 \right] d\varphi \quad (4)$$

$h_0 > 0$: unstable; $h_0 = 0$: semi-stable; $h_0 < 0$: stable.

5 Validated Examples

6 Key Findings

1. **First purely symbolic GHFP variant:** The introduction of composite Simpson quadrature into the GHFP framework makes the proposed method the first variant capable of **purely**

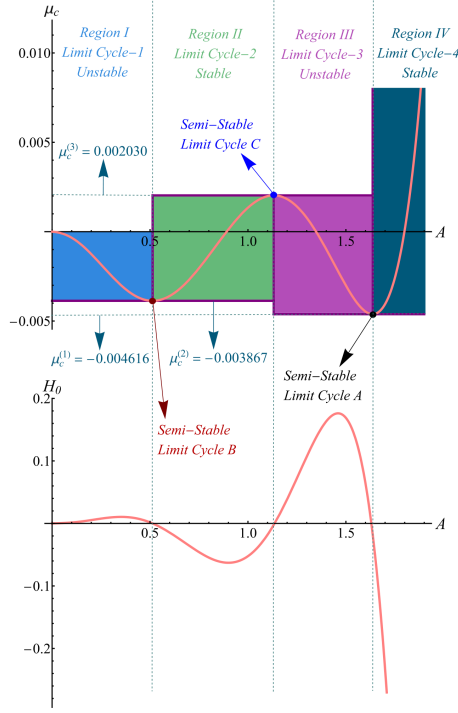


Figure 2: **Figure 2.** Evolution of limit cycles in the parametric plane (μ_c - A curve). The complete lifecycle is visible: emergence at the semi-stable bifurcation point, splitting into stable and unstable branches, and eventual termination.

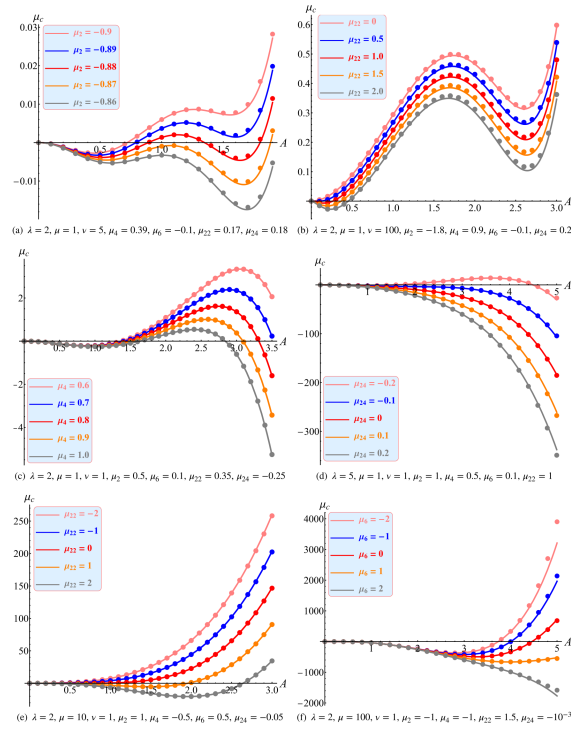


Figure 3: **Figure 5.** μ_c - A curves with different system parameters ($\varepsilon = 0.01$). Solid lines: MGHPF method. Dotted: Runge-Kutta reference. The method accurately captures limit cycle evolution across a wide parameter range.

symbolic execution for complicated oscillators — no parameter pre-assignment needed. This is the foundational innovation that enables all subsequent MGHFP work.

2. **Complete lifecycle from parametric plane:** The μ_c - A curve reveals the entire lifecycle of each limit cycle: from generation (semi-stable bifurcation), through evolution (stable and unstable branches), to termination (homoclinic bifurcation or collapse to equilibrium).
3. **Homoclinic and heteroclinic bifurcations predicted:** By setting the limit cycle amplitude equal to the saddle point coordinate in Eq. (3), both homoclinic and heteroclinic bifurcation thresholds are obtained analytically.
4. **Method validated on maximally challenging test case:** The DHRL oscillator combines Duffing stiffness, Harmonic irrational restoring force, and six-term generalized Rayleigh–Liénard damping — deliberately chosen to stress-test the method. All analytical predictions agree with fourth-order Runge–Kutta results.
5. **Foundation for the MGHFP series:** This paper establishes the MGHFP framework that is subsequently extended: to SD oscillators with quartic damping (IJNLM 2025, Padé enhancement), and to coupled SD oscillators with two irrational terms (Phys. Scr. 2026, the most complex application).

7 Engineering Significance

The generalized DHRL oscillator is not a contrived example — its damping structure (mixing velocity-dependent Rayleigh terms and displacement-dependent Liénard terms) models realistic scenarios in MEMS resonators with squeeze-film damping, vibration isolators with nonlinear electromagnetic damping, and self-excited systems in energy harvesting. The MGHFP method provides the first analytical tool capable of producing continuous amplitude–parameter curves for such systems.

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